Electrostatic Potential Energy Of System Of Point Charges:

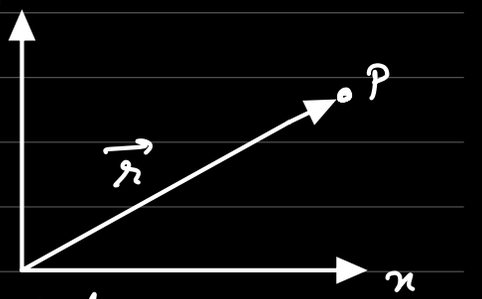
Recall that we defined potential energy as negative of work done by conservative forces in bringing a system from certain reference configuration to the given configuration. In same spirit we define electrostatic potential energy of a system of point charges as negative of work done in assembling the system of point charges bringing each one of them from infinity. (Reference configuration here is assumed to be the configuration of infinite separation).

Electric Potential: Just as concept of electric field is useful for calculation of electric forces on arbitrary charges placed at any point in space, similarly concept of electric potential is useful for calculating changes in electric potential energies of systems when an arbitrary charge is brought from infinity to certain point of space. Usually we define potential at an arbitrary point as follows:

"Negative of work done by the electric field in bringing of a charged particle from infinity to a given point of space per unit charge is called potential at that point of space."

Mathematically, if $\vec{E}(r)$ is electric field function and V potential at any point in space then:

$$V = -\int_{\infty}^{r} \vec{E} \cdot d\vec{r}$$



By inference this is also the change in potential Energy when a unit charge is brought from infinity to that given point. Thus if instead of a unit charge, a charge q were brought from infinity, the potential energy change would be:

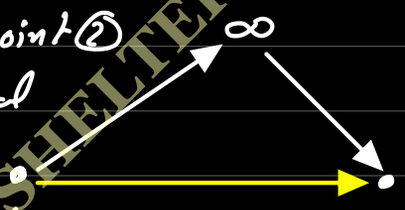
$\Delta U = qV$, similarly if a charge were removed to infinity then $\Delta U = -qV$ where V is potential at the point (excluding contribution of q itself).

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→ Only conventionally potential is defined making infinity as reference, though in some problems, some other points may be taken as reference for defining potential, thus definition of potential would read as "Negation of work done in bringing a charge from a reference point to the given configuration per unit charge is called potential at that point".

POTENTIAL DIFFERENCE: Let V_1 and V_2 be the potentials at two points of space then if we move from point 1 to point 2, then potential difference is simply $\Delta V = V_2 - V_1$.

If a charge q is moved from point ① to point ② it can be easily seen that change in potential energy is simply $q\Delta V$. This can be shown as follows. Since electrostatic force is a conservative force ∴ potential energy change should not depend on path followed. Thus instead of going from ① to ② directly we can route through ∞, i.e. taking the path ① → ∞ → ②.



Now as per our discussion in previous section, $\Delta U_{1\to\infty} = -qV_1$

Similarly $\Delta U_{\infty\to 2} = qV_2$ thus $\Delta U_{1\to 2} = \Delta U_{1\to\infty} + \Delta U_{\infty\to 2}$

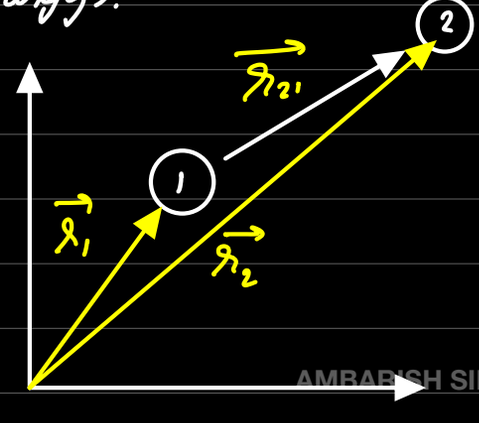
$$\Rightarrow \quad \Delta U = -qV_1 + qV_2 = q(V_2 - V_1) = q\Delta V.$$

$$\Rightarrow \quad \boxed{\Delta U = q\Delta V}$$

This gives us an alternative definition of potential difference as negative of work done by electric forces in going from an initial to final point per unit charge. (This will also be equal to work done by external force per unit charge in making this movement without changing the Kinetic Energy).

Mathematically:

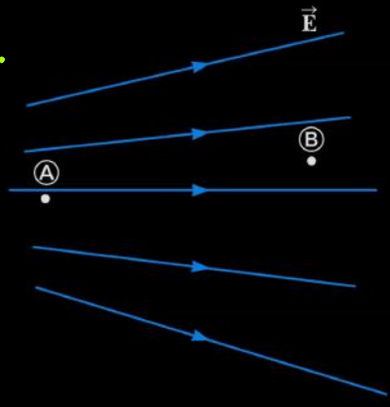
$$\boxed{V_2 - V_1 = \int_{\vec{r}_1}^{\vec{r}_2} -\vec{E}\cdot d\vec{r}}$$



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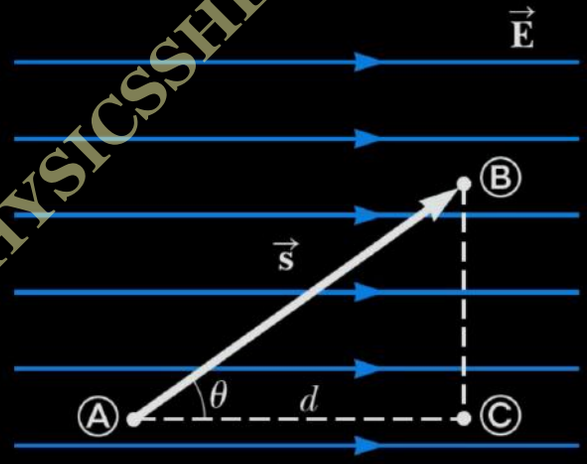
For elemental displacements we can also write elemental form of the above equation as $\boxed{dV = -\vec{E}\cdot d\vec{r}}$

→ Potential decreases while moving along the field lines while it increases while moving against the field lines. For instance in adjacent figure, $V_B < V_A$.



→ For positive charges potential energy decreases while moving along field lines while for negative charges, potential energy decreases while moving against the field lines. (Thus an electron at B has higher potential energy than an electron at A for the shown field).

Potential Difference In Uniform Field:



In going from A → B, work done by electric field per unit positive charge: $\Delta W = E \times S \times \cos\theta = Ed$

$\Rightarrow \Delta V = -\Delta W = -Ed$

$\Rightarrow V_B - V_A = -Ed$

(Note that only component of displacement along the field matters). You can readily see, that while moving from B to C no work is done by electric field, thus B and C are at same potential or equipotential.

→ Whenever we move along a path perpendicular to electric field lines, no work is done by electric field and such contours are called equipotential contours (or curves or line or surfaces in 3D space). For instance in above example, equipotentials are vertical lines, for point charges, equipotential surfaces are concentric spherical shells centred at the charge.

Example: Field in a certain region of space is given as $\vec{E}(x) = E_0 e^{-\alpha x}\hat{x}$. Find potential at a general point assuming zero potential at the origin.

Sol: $dV = -\vec{E}\cdot d\vec{x} = -E_0 e^{-\alpha x}\hat{x}\cdot(dx\,\hat{x})$



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$$\Rightarrow \quad dV = -E_0 e^{-\alpha x} dx \quad \Rightarrow \int_0^V dV = \int_0^x -E_0 e^{-\alpha x} dx$$

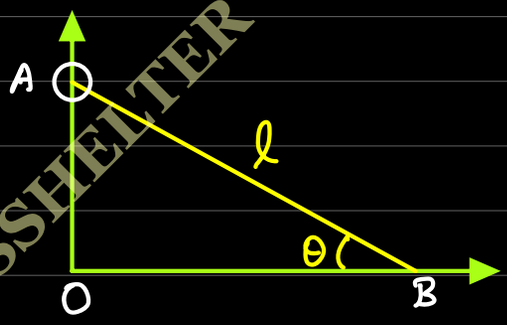
$$\Rightarrow V = \frac{E_0}{\alpha}\left[e^{-\alpha x} - 1\right] \text{ Ans.}$$

Think: Suppose you release a positive point charge q from origin, what will be its KE when it reaches ∞.

Sol: Loss in PE = Gain in KE $\Rightarrow \Delta U + \Delta KE = 0 \Rightarrow \Delta KE = -\Delta U = -q[V(\infty) - V(0)]$

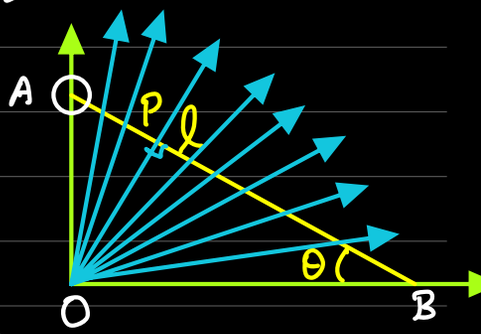
$$\Rightarrow \Delta KE = -q[-1 - 0)\frac{E_0}{\alpha} = \frac{qE_0}{\alpha} \text{ Ans.}$$

EXAMPLE: A string is tied between points A and B on coordinate axes and a bead carrying charge q is threaded on it. The space carries a field $\vec{E} = E_0 \hat{x}$. Calculate the minimum velocity to be given to the bead along thread so that it may reach B. (mass of bead is m). (Ignore gravity)



Solution: For solving such questions minimum initial KE is governed by maximum potential energy to be overcome in the process and not just difference in final and initial PE. (Think of a potential hill barrier that we discussed when we studied equilibria).

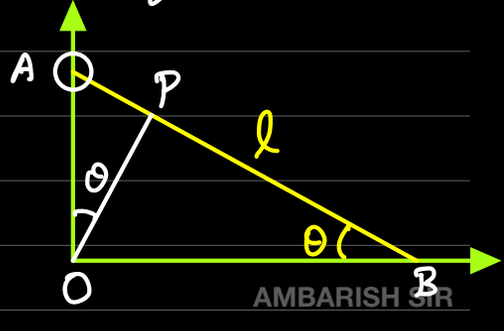
Here we see that maximum potential energy will occur at point P because before this point electric field would be doing negative work ∵ angle between field and displacement would be obtuse. So our aim is only to overcome point P.



Now $E = E_0 \hat{i}$, $\Rightarrow \quad dV = -\vec{E} \cdot d\vec{x} = -E_0 \hat{i} \cdot (dx\,\hat{i})$

$\Rightarrow dV = -E_0 dx \Rightarrow V = -E_0 x + C$, we can conveniently chose constant to be zero, thus $V = -E_0 x$. Thus maximum potential occurs when x is min, i.e. $\perp$ point from origin.

Now $OA = l\sin\theta$ & $OP = OA\cos\theta$, $\therefore OP = l\sin\theta\cos\theta$.



Now $V_i = -E_0 \times OA = -E_0 l\sin\theta$

$V_P = -E_0 \times OP = -E_0 l\sin\theta\cos\theta$

Thus $\Delta U_{A\to P} = q(V_P - V_A) = qE_0 l\sin\theta(1 - \cos\theta)$

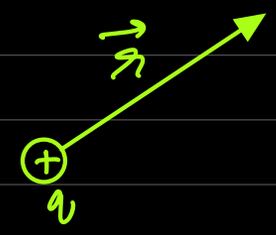
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Thus in limiting case, by energy conservation:

$$qE_0\ell(1-\cos\theta)\sin\theta = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2qE_0\ell(1-\cos\theta)\sin\theta}{m}}\ \text{ans.}$$

Potential Due To A Point Charge:

For a point charge: $\vec{E} = \frac{kq}{r^2}\hat{r}$

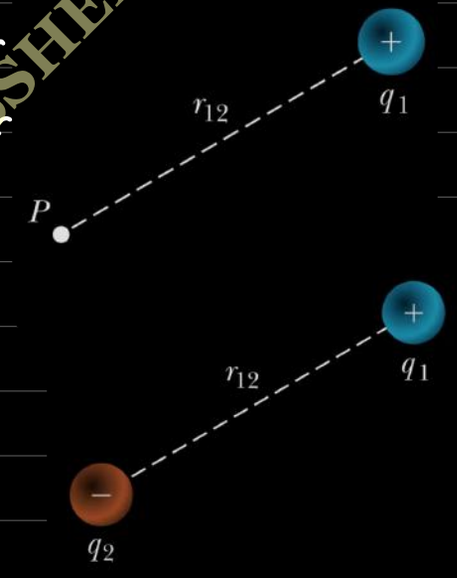


$$\Rightarrow dV = -\vec{E}\cdot d\vec{r} = -\frac{kq}{r^2}\hat{r}\cdot(dr\,\hat{r}) = -\frac{kq}{r^2}dr$$

Now $\int_0^V dV = \int_\infty^r -\frac{kq}{r^2}dr \Rightarrow \boxed{V = \frac{kq}{r}}$

Potential Energy Of A Pair Of Particles:

We wish to find potential energy of a system of two particles, i.e negative of work done by electric force in assembling this system bringing the particles from ∞. Imagine the particles to be at ∞ separation initially, now to bring the first charge we don't need to do any work ∵ other charge exerts no force due to infinite separation. Now bringing the second charge at r_{12} will require some work because ① is exerting force on ②. Potential at P due to ① is $V_{21} = \frac{kq_1}{r_{12}}$ thus OU upon bringing q_2 = OU = $q_2 \times V_{21}$



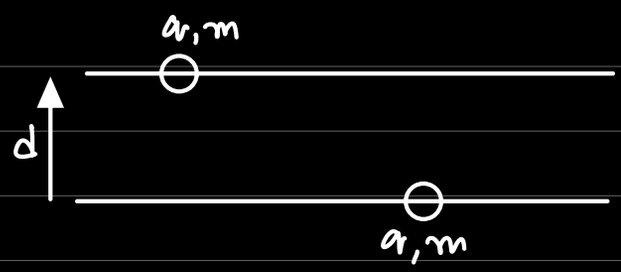
$$\Rightarrow \boxed{U = \frac{kq_1q_2}{r_{12}}}$$

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Example: (Pathfinder Build-48):

48. Two identical beads each of mass m and charge q can slide on two long parallel insulating frictionless rigid rods that are a distance d apart. Initially the beads are so far from each other that no electrostatic interaction exists between them. With what speed must one of the beads be projected so that it passes over the other bead? Assume radii of the beads negligibly small as compared to the distance between the rods. Neglect effects of gravity.

Solution: Maximum potential Energy of the system occurs when beads are nearest to each other i.e at a distance d (beads need to overcome this shortest separation before carrying on their escape to ∞). We can work in CM frame with reduced mass. (Alternatively we could use simultaneous momentum and energy conservation). Thus in the limiting case:



$$\frac{1}{2}\mu v^2 = \frac{kq^2}{d} \Rightarrow v = q\sqrt{\frac{2k}{\mu d}} = q\sqrt{\frac{4k}{md}}$$

$$\Rightarrow v > q/\sqrt{\pi\epsilon_0 m d} \quad \text{Ans.}$$

FINDING POTENTIAL BY SUPERPOSITION: Recall that potential at any point is given as:

$$V = \int_{\vec{r}_{ref}}^{\vec{r}} -\vec{E}\cdot d\vec{s} \quad \text{where } \vec{r}_{ref} \text{ is the}$$

reference position where potential is assumed to be zero. Now let us have a system of several particles generating fields $\vec{E}_1, \vec{E}_2 \cdots \vec{E}_n$ such that net field at any point in space is $\vec{E} = \vec{E}_1 + \vec{E}_2 + \cdots \vec{E}_n$, then potential at any point is:

$$V = \int_{\vec{r}_{ref}}^{\vec{r}} -\vec{E}\cdot d\vec{s} = -\int_{\vec{r}_{ref}}^{\vec{r}} (\vec{E}_1 + \vec{E}_2 + \cdots \vec{E}_n)\cdot d\vec{s}$$

Using distribution property of dot product we can write this as:

$$V = -\int \vec{E}_1\cdot d\vec{s} - \int \vec{E}_2\cdot d\vec{s} + \cdots - \int \vec{E}_n\cdot d\vec{s} = V_1 + V_2 + V_3 + \cdots V_n.$$

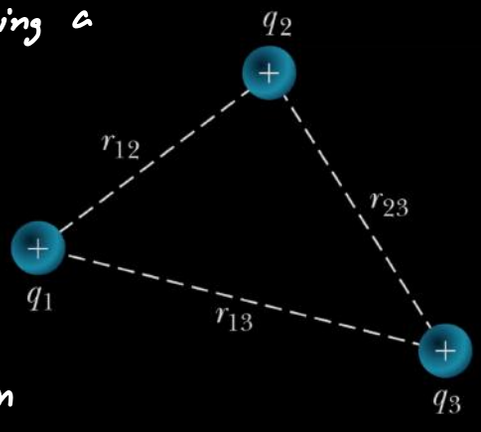
Thus potential due to system of point charges follows superposition principle. We can simply add the potential contribution of each charge individually as if other charges were not present.

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Potential Energy Of A System Of Three Point Charges:

Recall from previous section that change in PE in assembling a system of two point charges is

$$\Delta U_1 = \frac{k q_1 q_2}{r_{12}}.$$



Now suppose we bring the third charge from infinity under influence of fields due to these two charges as shown. By superposition potential at location of q_3 due to these two charges is given as:

$$V_3 = \frac{k q_1}{r_{13}} + \frac{k q_2}{r_{23}} \Rightarrow \Delta U \text{ in this step is given as:}$$

$$\Delta U_2 = q_3 V_3 = \frac{k q_1 q_3}{r_{13}} + \frac{k q_2 q_3}{r_{23}}.$$ Thus for the entire process in the entire process is, the total potential energy is the sum of potential energies involved in two steps. Thus:

$$U = \Delta U = \Delta U_1 + \Delta U_2 = \frac{k q_1 q_2}{r_{12}} + \frac{k q_2 q_3}{r_{23}} + \frac{k q_1 q_3}{r_{13}}$$

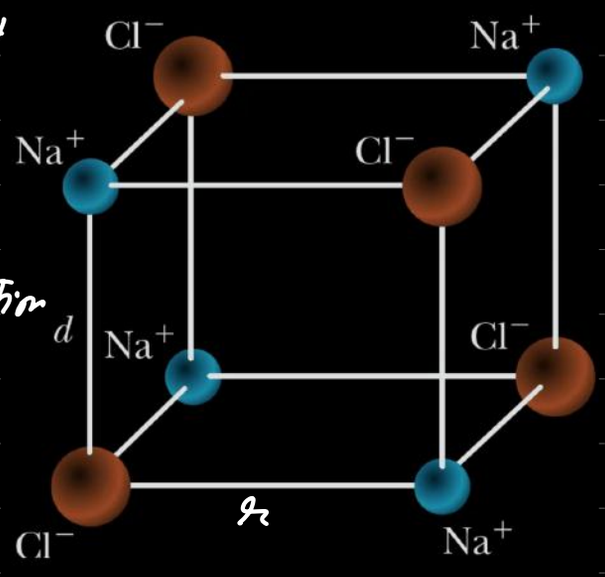
It can be readily seen that if now a fourth charge is brought it will have an interaction with existing three charges and thus add three more terms to the total PE to the system.
From this pattern it is clear that PE of a system of n particles is given as:

$$\boxed{U = \sum \frac{k q_i q_j}{r_{ij}}}$$

Here summation is to be carried out for each pairwise interaction thus total number of terms will be $C(n, 2)$. Note that if r_{ij} term is counted don't again count r_{ji} term. Each interaction is to be counted only once.

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EXAMPLE: Adjacent figure shows arrangement of Na^+ and Cl^- ions in a cube like structure. Assuming edge of cube to be r and Na^+ charge as $+q$ and Cl^- charge as $-q$, Calculate the interaction energy of the structure.



Solution: To calculate interaction energy we must systematically count the number of interactions:

① Along edges $[q, -q, r] \times 12$

② Along face diagonals $(q \times q$ or $(-q \times -q), r\sqrt{2}) \times 12$

③ Along Body Diagonals $(q, -q, r\sqrt{3}) \times 4$.

Thus interaction energy:

$$-\frac{kq^2}{r} \times 12 + \frac{kq^2}{r\sqrt{2}} \times 12 - \frac{kq^2}{r\sqrt{3}} \times 4 = \frac{kq^2}{r}\left[-12 + \frac{12}{\sqrt{2}} - \frac{4}{\sqrt{3}}\right]$$

$$\approx -5.82 \frac{kq^2}{r} \quad \text{Ans.}$$

→ Thus by forming a crystall lattice, potential Energy is lowered (when at ∞, potential energy was zero, but now it is negative), this explains why crystals are formed.

Think yourself: How much energy would be required to remove one of the ions from the above structure.

Sol: Suppose we remove a Cl^-, this was involved in three face diagonals, three edges and one body diagonal, thus these interaction will not exist any more, accordingly new U:

$$-\frac{kq^2}{r} \times 9 + \frac{kq^2}{r\sqrt{2}} \times 9 - \frac{kq^2}{r\sqrt{3}} \times 3 = \frac{kq^2}{r}\left[-9 + \frac{9}{\sqrt{2}} - \sqrt{3}\right] \approx -4.4 \frac{kq^2}{r}$$

Thus potential energy is increased, now $\Delta W_{ext} = \Delta U$

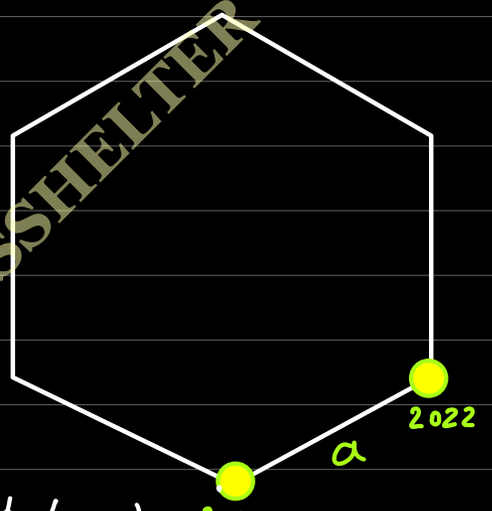
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$\Rightarrow$ $\Delta W = (-4.4 + 5.8)\frac{kq^2}{a} = \frac{1.4kq^2}{a}$. (This much work is required to remove a Cl^-, same amount of work will be required to remove an Na^+ as well.).

EXAMPLE : KROTOV-3.77.

Small identical balls with equal charges are fixed at the vertices of a right 2023-gon with side a. At a certain instant, one of the balls is released, and a sufficiently long time interval later, the ball adjacent to the first released ball is freed. The kinetic energies of the released balls are found to differ by K at a sufficiently long distance from the polygon. Determine the charge q of each ball.

Solution: For ease of visualisation a representative figure is made. When the first charge is released it loses its interaction energy with other 2021 charges + interaction energy with 2022nd charge, while when second charge is released it only loses its interaction energy with 2021 charges (which part was just like that of the first charge). Thus difference in their KE is due to the one additional interaction which was present for 2023rd charge but absent for 2022nd charge.

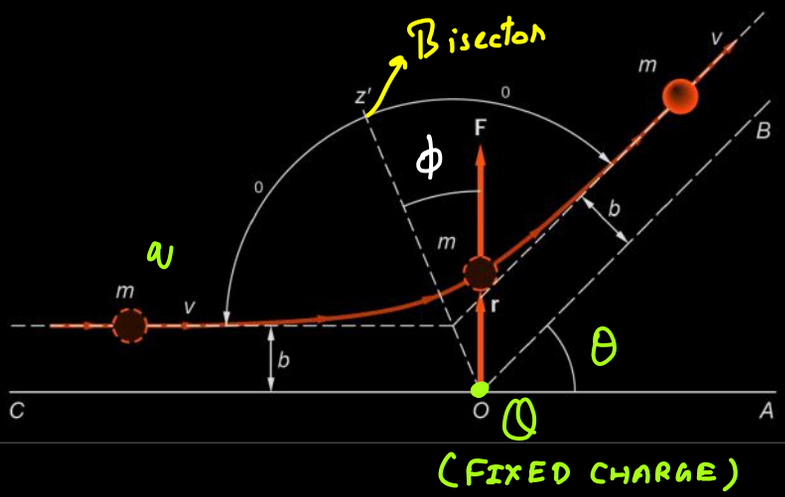


$\therefore K = \frac{kq^2}{a} \Rightarrow q = \sqrt{\frac{Ka}{k}} = \sqrt{4\pi\epsilon_0 K a}$ Ans

EXAMPLE : RUTHERFORD SCATTERING :

A particle of mass m and charge q is thrown from infinity with an impact parameter b and Kinetic Energy T as shown in the figure prove that if θ is the angle of deviation, then:

$$\boxed{\tan(\theta/2) = \frac{kQq}{2Tb}}$$



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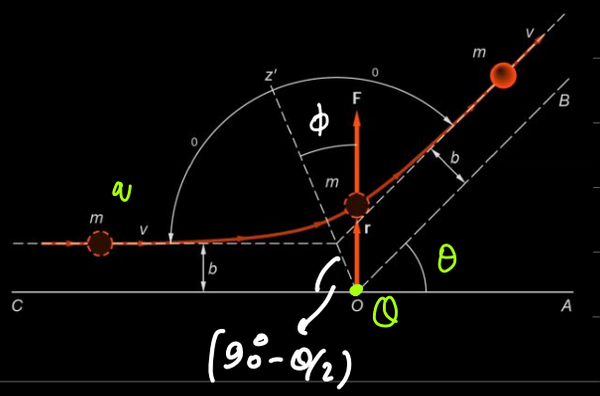
Solution: Linear momentum change occurs along the bisector of the initial and final directions

$$\int \frac{kqQ}{r^2}\cos\phi\, dt = 2mv\cos(90^\circ - \theta/2) \quad (1)$$

$$\Rightarrow \int (kqQ/r^2)\cos\phi\, dt = 2mv\sin(\theta/2) \quad (2)$$

Also by angular momentum conservation:

$$mvb = mr^2(d\phi/dt) \Rightarrow dt = \frac{r^2 d\phi}{bv} \quad (3)$$



Using (3) in (1)

$$\int_{-(90^\circ-\theta/2)}^{+(90^\circ-\theta/2)} \frac{kqQ\cos\phi}{r^2}\cdot\frac{r^2}{bv}\, d\phi = 2mv\sin(\theta/2)$$

$$\Rightarrow \frac{kQq}{bv}\left[\sin\phi\right]_{-(90^\circ-\theta/2)}^{+(90^\circ-\theta/2)} = 2mv\sin(\theta/2)$$

$$\Rightarrow \frac{kQq}{bv}\cdot 2\cos(\theta/2) = 2mv\sin(\theta/2) \Rightarrow \tan(\theta/2) = \frac{kQq}{2\times\frac{1}{2}mv^2 b}$$

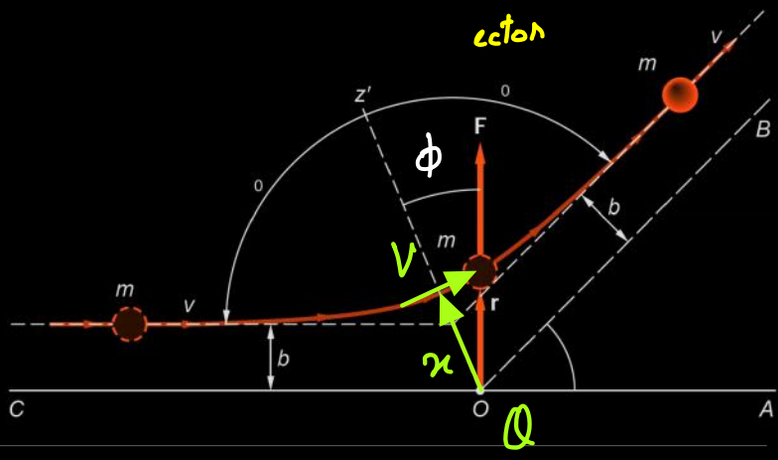
$$\Rightarrow \boxed{\tan(\theta/2) = \frac{kQq}{2Tb}} \quad \text{QED.}$$

EXAMPLE: What is the distance of closest approach in Rutherford Scattering?

Solution: Let r be the distance of closest approach and corresponding speed be V then:

By Energy Conservation:

$$\frac{1}{2}mv^2 = \frac{1}{2}mV^2 + \frac{kQq}{r} \quad (1)$$



By angular momentum Conservation:

$$mvb = mVr \quad (2)$$

Solving (1) & (2):

$$r = \frac{kQq}{mv^2}\left(1+\sqrt{1+\left(\frac{bmv^2}{kQq}\right)^2}\right)$$

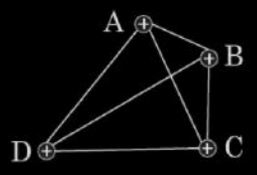
$$\Rightarrow r = b\tan(\theta/2)\left(1+\csc(\theta/2)\right) = b\left(\tan(\theta/2)+\sec(\theta/2)\right)$$

$$\Rightarrow \boxed{r = b\left(\tan(\theta/2)+\sec(\theta/2)\right)} \quad \text{Ans.}$$

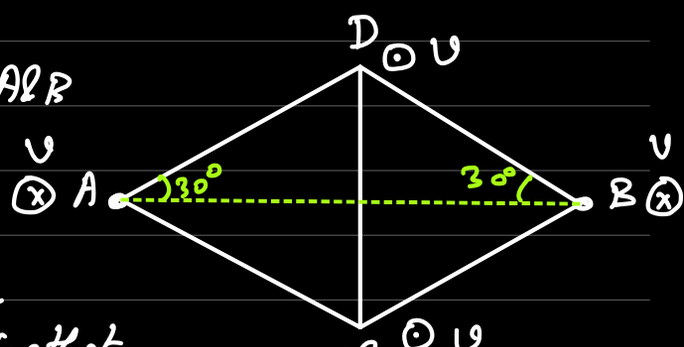
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EXAMPLE: PATHFINDER BUILD 50

Four identical particles A, B, C and D each of mass m and charge q are connected by insulating light inextensible threads of equal length l to make a tetrahedron as shown in the figure.



The thread connecting particles A and B is cut. Find maximum speed acquired by every particle in subsequent motion.



Solution: Here only distance between points A & B is changing while all other distances are maintained, thus PE loss is only the loss in interaction energy between A & B. Maximum loss occurs when A & B are fastest i.e a planar rhombus is formed. Here for momentum conservation, the inward speed of A & B must be equal to outward speed of C & D.

Now loss in PE = Gain in KE,

$$\left(\frac{2q^2}{l} - \frac{2q^2}{2l\cos 30^\circ}\right) = \frac{1}{2}mv^2 \times 4 \Rightarrow v = \sqrt{\frac{q^2}{8\pi\varepsilon_0 ml}\left(1 - \frac{1}{\sqrt{3}}\right)}\ \text{Ans.}$$

FINDING FIELD FUNCTION FROM POTENTIAL FUNCTION :

Recall the differential form of potential equation, $dV = -\vec{E}\cdot d\vec{r}$

Let $\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$, Now suppose if I move // to x axis keeping y & z coordinates fixed by some elemental displacement given as $d\vec{r} = dx\,\hat{i}$, then: $dV = -\vec{E}\cdot d\vec{r}$

$\Rightarrow dV = -E_x dx \Rightarrow E_x = -\frac{dV}{dx}$, but since V is a 3D function here, actually by $\frac{dV}{dx}$ we mean here partial derivation

$\because$ We kept y & z fixed. Thus:

$$\boxed{E_x = -\partial V/\partial x}$$ Similarly:

$$\boxed{E_y = -\partial V/\partial y}$$ and

$$\boxed{E_z = -\partial V/\partial z}$$

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Thus we can also write $\vec{E} = -\left[\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right]$

or in compact form $\boxed{\vec{E} = -\vec{\nabla} V}$ where $\vec{\nabla}$ is an operator defined as

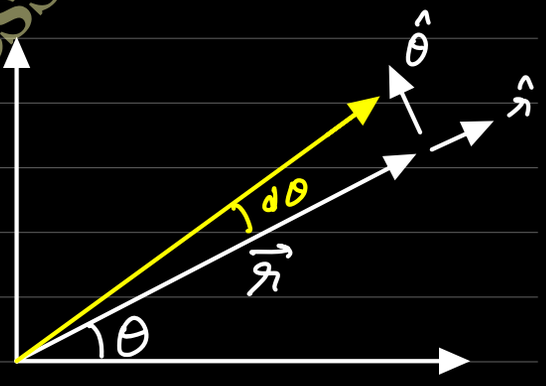
$$\boxed{\vec{\nabla} = \left[\hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}\right]}$$

EXAMPLE : Potential in a certain region of space is given as $V = \alpha x^2 + \beta y^2 + \gamma z^2$, find the field function in the region.

Solution: $\vec{E} = -\vec{\nabla} V = -\left[2\alpha x\,\hat{i} + 2\beta y\,\hat{j} + 2\gamma z\,\hat{k}\right]$ Ans.

FINDING FIELD FROM POTENTIAL FUNCTION IN POLAR 2D COORDINATES

Recall that in cartesian coordinates to find field from potential we had found rate of change of potential wrt movement in that direction (keeping no movement in other directions). Same idea can be extended to polar coordinates, thus if we move only radial direction keeping θ coordinate fixed, we will get radial component of electric field as:



$$\boxed{E_r = -\frac{\partial V}{\partial r}}$$

Similarly if we move a small distance $r\,d\theta$ in tangential direction keeping r fixed, we get $E_\theta = -\frac{dV}{r\,d\theta}$, however since V is a two dimensional function, here by $dV/d\theta$ we mean $\partial V/\partial\theta$, $\therefore$

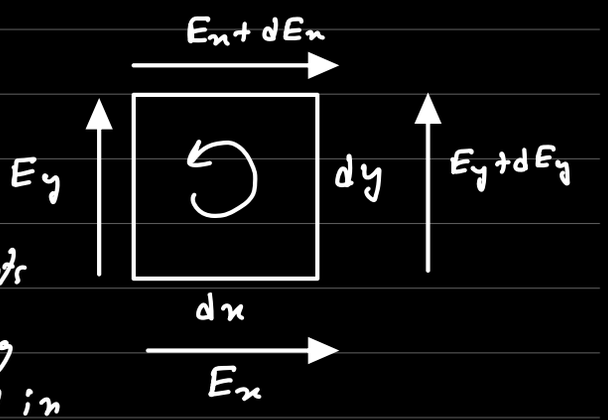
$$\boxed{E_\theta = \frac{-1}{r}\frac{\partial V}{\partial \theta}}$$

EXAMPLE : If $V = \frac{a^2}{r^2} V_0 \sin\theta$, Find E_r and E_θ in the plane.

Solution: $E_r = -\partial V/\partial r = -\left(-2a^2/r^3\right) V_0 \sin\theta$, $E_\theta = \frac{-1}{r}\frac{\partial V}{\partial \theta} = \frac{-a^2}{r^3} V_0 \cos\theta$ Ans.

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Test Of A Conservative Field:



For a conservative field, work done in a closed path must be zero. Consider a small rectangular loop of dimensions $dx \times dy$ as shown, with field components along its sides as shown. Now Travelling along the loop, work done by Electric field in closed loop is given as:

$$dW = E_x\,dx + (E_y + dE_y)dy - (E_x + dE_x)dx - E_y\,dy$$

This must be zero for field to be conservative, i.e:

$$dE_y\,dy - dE_x\,dx = 0 \implies \frac{dE_y}{dx} = \frac{dE_x}{dy}$$

However since E_x and E_y themselves are multivariable functions, $\therefore$ here by dE_x/dy we actually mean $\partial E_x/\partial y$. Thus for conservative fields:

$$\boxed{\begin{aligned} \partial E_x/\partial y &= \partial E_y/\partial x \\ \partial E_y/\partial z &= \partial E_z/\partial y \\ \partial E_z/\partial x &= \partial E_x/\partial z \end{aligned}}$$

Similarly and

All these three conditions can be written compactly using ∇ operator as :

For Conservative fields $\boxed{\vec{\nabla} \times \vec{E} = 0}$, This can be readily verified using matrix method for cross products.

$$\vec{\nabla} \times \vec{E} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ E_x & E_y & E_z \end{vmatrix} = \begin{aligned} &\hat{i}\,(\partial E_z/\partial y - \partial E_y/\partial z) \\ &-\hat{j}\,(\partial E_z/\partial x - \partial E_x/\partial z) \\ &+\hat{k}\,(\partial E_y/\partial x - \partial E_x/\partial y) = 0 \end{aligned}$$

($\because$ Cross derivatives are equal).

→ Note that for conservative field $\vec{\nabla} \times \vec{E}$ should be identically zero and not just zero at some particular points.

→ For applying $\vec{\nabla} \times \vec{E}$ test, the function must be continuous and differentiable, otherwise we may end up in a fallacy.

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→ Exception Alert: The field $\frac{1}{(x^2+y^2)}[y\hat{i} - x\hat{j}]$, might appear to satisfy the test of conservation fields on a mechanical checking of cross derivatives, however the function is not defined at origin. It turns out that for any path going around the origin the work done will be non zero.

A SPECIAL FIELD $\vec{E} = (y\hat{i} - x\hat{j})/(x^2+y^2)$:

$$E_x = \frac{y}{(x^2+y^2)} \Rightarrow \frac{\partial E_x}{\partial y} = \left(\frac{(x^2+y^2)\times 1 - y\times 2y}{(x^2+y^2)^2}\right)$$

$$\Rightarrow \frac{\partial E_x}{\partial y} = \frac{x^2-y^2}{(x^2+y^2)^2} \quad (1)$$

$$E_y = \frac{-x}{x^2+y^2} \Rightarrow \frac{\partial E_y}{\partial x} = -\left[\frac{(x^2+y^2)[1] - (x\cdot 2x)}{(x^2+y^2)^2}\right]$$

$$\Rightarrow \frac{\partial E_y}{\partial x} = \frac{x^2-y^2}{(x^2+y^2)^2} \quad (2)$$

From (1) & (2) it appears $\frac{\partial E_y}{\partial x} = \frac{\partial E_x}{\partial y}$ ∴ appears Conservative.

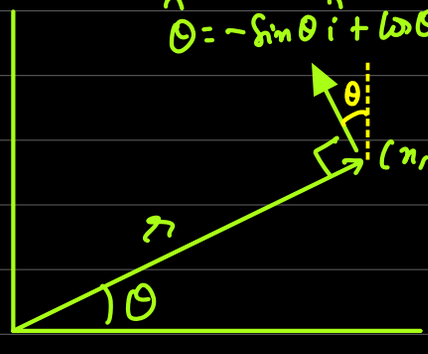
But Hold Your Horses.... !!

Let us convert the field to polar form :

$x = r\cos\theta \quad y = r\sin\theta$

$$\Rightarrow \vec{E} = \frac{1}{x^2+y^2}[y\hat{i} - x\hat{j}]$$

$$= \frac{1}{r^2}[r\sin\theta\,\hat{i} - r\cos\theta\,\hat{j}]$$



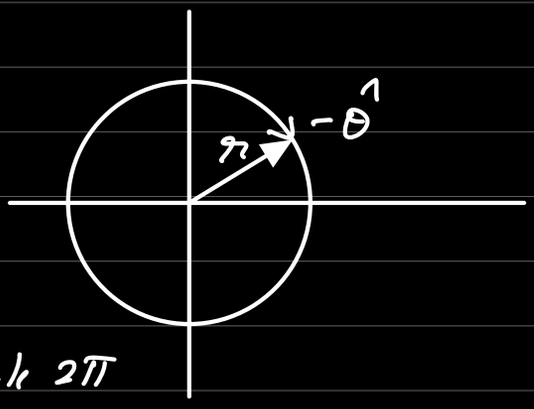
$\vec{E} = \frac{-1}{r}\hat{\theta}$, This is always along clockwise tangent thus field will form complete circular field line at any r

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A general displacement in the 2D plane is given by $d\vec{s} = dr\,\hat{r} + r\,d\theta\,\hat{\theta}$

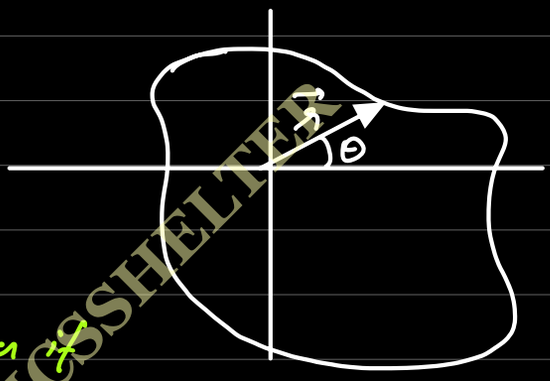
$\therefore$ elemental work done by this force is given as:

$$dW = \vec{E}\cdot d\vec{s} = \left(-\frac{1}{r}\hat{\theta}\right)\cdot\left[dr\,\hat{r} + r\,d\theta\,\hat{\theta}\right]$$



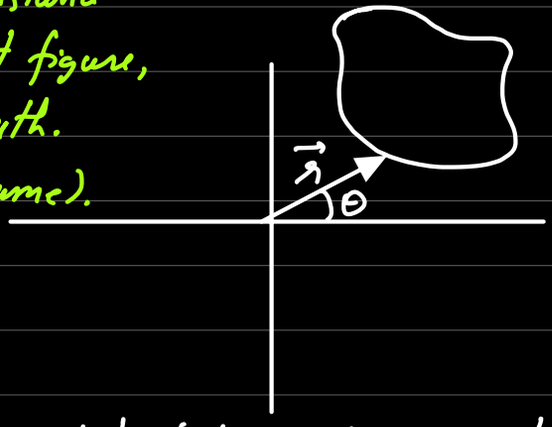
$\Rightarrow dW = -d\theta$, thus in going an angle 2π around the origin work done is:

$$W = \int_0^{2\pi} -d\theta = -2\pi$$



→ Note that the curve could be arbitrary (i.e r could vary randomly), however if curve goes completely around origin, work done in closed path is non zero, $\therefore$ THE FIELD IS NON-CONSERVATIVE!!!

→ So long as you create a closed path not going around the origin, work done is zero. For instance for the path shown in the adjacent figure, work done is zero in the closed path. (Because both integration limits of θ are same).



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FINDING POTENTIAL FUNCTION FROM FIELD:

Recall that for finding field function from potential function, we had resorted to partial differentiation, accordingly for finding potential function from the field function we will have to do the partial Integration. For a two dimensional field, partial Integration wrt x would involve adding an arbitrary function of y because $\frac{\partial}{\partial x}(f(y)) = 0$ irrespective of form of $f(y)$. Similarly for a three dimensional field, partial integration wrt x would involve addition of an arbitrary function of y and z. $\because \frac{\partial}{\partial x} f(y,z) = 0$ irrespective of form of $f(y,z)$.

EXAMPLE: Find potential function if the field function is given as $\vec{E} = 2(x+y)\hat{i} + 2(x+y)\hat{j}$

Solution:

$$\partial V/\partial x = -E_x = -2(x+y) \text{ ①} \qquad \partial V/\partial y = -E_y = -2(x+y) \text{ ②}$$

From ①, $V = -(x^2+2xy) + f(y)$ ③ substituting this in LHS of ②

$$\frac{\partial V}{\partial y} = -2x + \frac{df}{dy} = -2x-2y \Rightarrow \frac{df}{dy} = -2y \Rightarrow f = -y^2 + C \text{ ④}$$

Now putting ④ in ③, $V = -(x^2+2xy+y^2) + C$ Ans.

→ Choice of constant can be arbitrary, because it depends upon where we have taken reference potential to be zero. For instance in this example if we assume potential at origin to be zero then $C = 0$.

TRY YOURSELF: Electric field in a certain region of space is given as $\vec{E} = a(x\hat{i} + y\hat{j})/(x^2+y^2)$. Compute the potential function, Take reference potential as zero at $(1/\sqrt{2}, 1/\sqrt{2})$.

Solution:

$$\partial V/\partial x = -ax/(x^2+y^2) \text{ ①} \qquad \partial V/\partial y = -ay/(x^2+y^2) \text{ ②}$$

From ①:

$$V = -\frac{a}{2}\ln(x^2+y^2) + f(y) \text{ ③}$$

$$\Rightarrow \frac{\partial V}{\partial y} = \frac{-ay}{x^2+y^2} + \frac{df}{dy} = \frac{-ay}{(x^2+y^2)} \Rightarrow \frac{df}{dy} = 0 \Rightarrow f(y) = C.$$

$$\Rightarrow V = \frac{-a}{2}\ln(x^2+y^2) + C, \text{ Now } V(1/\sqrt{2}, 1/\sqrt{2}) = 0 \Rightarrow C = 0$$

$$\Rightarrow V = \frac{-a}{2}\ln(x^2+y^2) \text{ Ans.}$$

→ A curve with same potential everywhere is called an equipotential curve (It must be ⊥ to electric field lines everywhere).

→ Once we have potential function, we also have the family of equipotential curves as $V(x,y) = \text{Constant}$. For instance in this example, family of equipotential curves is $\frac{-a}{2}\ln(x^2+y^2) = C$

$\Rightarrow x^2+y^2 = C$, which is a family of circles.

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FINDING FAMILY OF EQUIPOTENTIAL CURVES FROM FIELD FUNCTION:

Recall from our discussion on field lines that slope of a field line for a field $E_x\hat{i} + E_y\hat{j}$ is given as $dy/dx = E_y/E_x$.
Now since equipotentials must be perpendicular to field lines (∵ only if we move ⊥ to field lines will the work done be zero).
∴ For equipotential curve family $\frac{dy}{dx} = -E_x/E_y$

∴ For orthogonal curve families, products of slopes must be -1.

EXAMPLE: Find the family of equipotential curves if the field function is given as: $\vec{E} = a(x\hat{i} + y\hat{j})/(x^2+y^2)$.

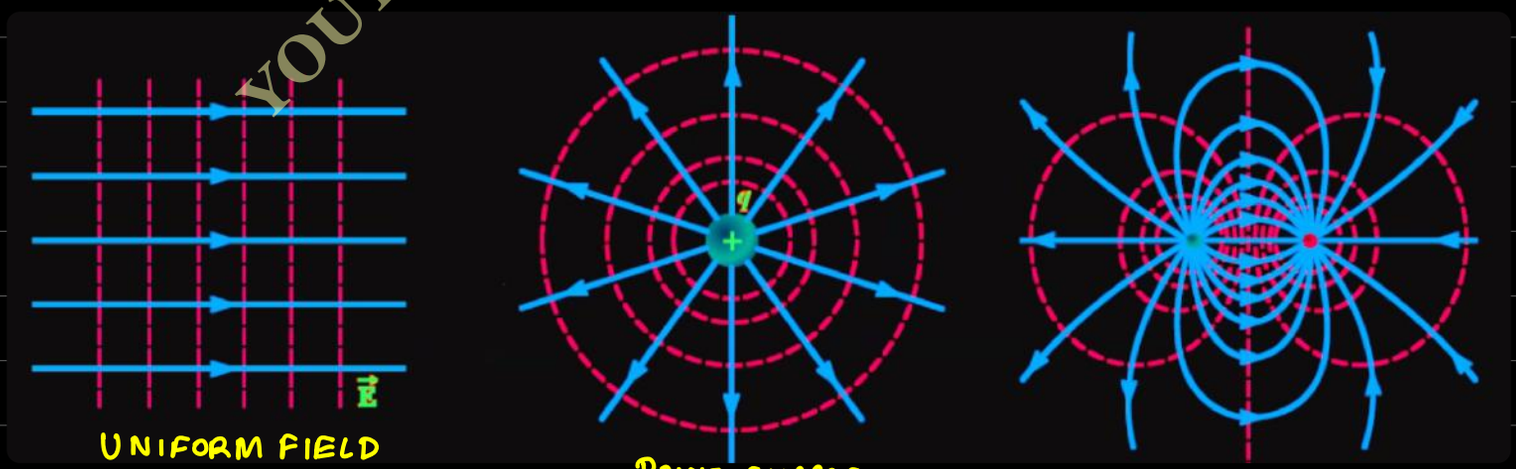
Solution: $E_x = ax/(x^2+y^2)$, $E_y = ay/(x^2+y^2)$

⇒ For family of equipotential curves, $\frac{dy}{dx} = -\frac{E_x}{E_y} = -\frac{x}{y}$

$\Rightarrow y\,dy = -x\,dx \Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} + C$ or $x^2 + y^2 = C$

→ This is the same family of curves that we obtained earlier by calculating potential function. Though this method successfully gives us the family of equipotential, it is useless for finding the exact potential function. While previous method gave us the potential function as well as equipotential family.

EXAMPLES OF EQUIPOTENTIAL SURFACES:



UNIFORM FIELD (PARALLEL PLANES) | POINT CHARGE (CONCENTRIC SPHERES) | DIPOLE

→ Instead of field lines, a field can also be visualised through equipotential surfaces. The density of equipotential surfaces

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is proportional to strength of the field and their normal in the direction of decreasing potential gives the direction of local electric field.

FINDING POTENTIAL FUNCTION FROM 3D FIELD FUNCTION:

Just as we had to do two integrations for finding a two dimensional potential function, we will have to do three integrations for finding a 3D potential function. (First two partial Integrations).

EXAMPLE: Field function in a region of space is given as: $\vec{E} = -[\hat{i} + z\hat{j} + (y+2z)\hat{k}]$, find Potential function in the space assuming that reference potential is zero at origin.

Solution: $\frac{\partial V}{\partial x} = 1$ ①, $\frac{\partial V}{\partial y} = z$ ②, $\frac{\partial V}{\partial z} = (y+2z)$ ③

① $\Rightarrow V = x + f(y,z)$, putting this in LHS of ②:

$\frac{\partial f}{\partial y} = z \Rightarrow f = yz + g(z) \Rightarrow$ Potential function is of the form $x + yz + g(z)$, putting in LHS of ③,

$y + \frac{dg}{dz} = y + 2z \Rightarrow \frac{dg}{dz} = 2z \Rightarrow g = z^2 + c$

$\Rightarrow V(x,y,z) = x + yz + z^2 + c$, Now $V(0,0,0) = 0$ (Given)

$\therefore V = x + yz + z^2$ Ans.

THINK YOURSELF: $\vec{E} = -[\cos x\,\hat{i} + z\cos yz\,\hat{j} + (y\cos yz + 2z)\hat{k}]$

the potential function for this field function, Given $V(0,0,0) = 0$.

Ans: $V(x,y,z) = \sin(x) + \sin yz + z^2$

Solution: $\frac{\partial V}{\partial x} = \cos(x) \Rightarrow V = \sin(x) + f(y,z)$ using for E_y:

$\frac{\partial f}{\partial y} = z\cos(yz) \Rightarrow f = \sin(yz) + g(z)$

$\Rightarrow V = \sin(x) + \sin(yz) + g(z)$, using in E_z:

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$$y\cos yz + \frac{dg}{dz} = y\cos yz + 2z \Rightarrow g = z^2 + C$$

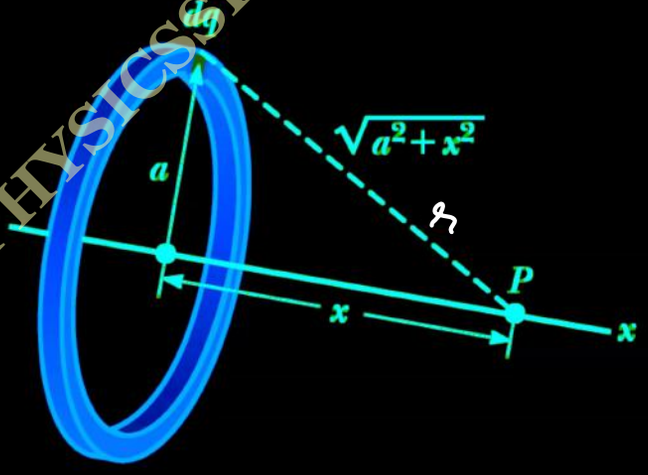
$\Rightarrow V = \sin(x) + \sin(yz) + z^2 + C$, Now $V(0,0,0) = 0 \Rightarrow C = 0$

$\Rightarrow V = \sin(x) + \sin(yz) + z^2$ ohm.

Potential Due To Continuous Charge Distributions: As discussed earlier, potential also follows superposition principle $\therefore$ potential of continuous charge distributions can be found by integrating the potential contributions due to it's various elements.

i.e $V = \int dV$.

Potential Due To A Uniformly Charged Ring:



Consider a point P on axis of a thin charged ring having a charge q and radius a. Due to a charge element, Potential contribution is given as:

$$dV = \frac{k\,dq}{r}$$

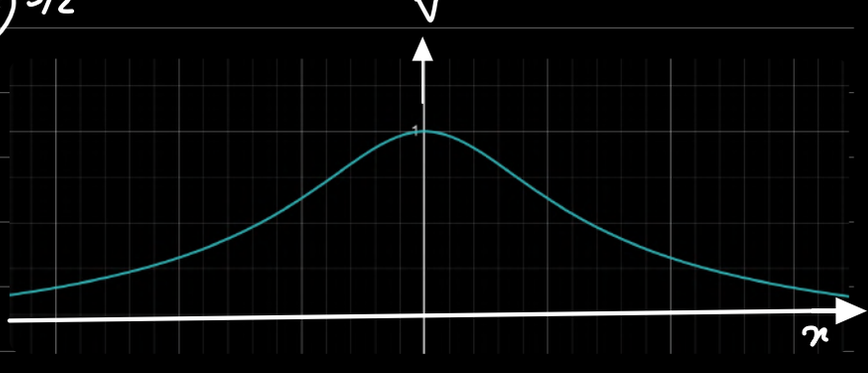
$$\Rightarrow V = \frac{k}{r}\int dq = \frac{kq}{r} = \frac{kq}{\sqrt{a^2+x^2}}$$

Thus $\boxed{V = \frac{kq}{\sqrt{a^2+x^2}}}$

→ One can readily find the axial field by differentiating the potential

i.e $E_x = -\frac{\partial V}{\partial x} = \frac{kqx}{(a^2+x^2)^{3/2}}$.

Plot of V vs x for ring:



For $x << a$, $V \approx \frac{kq}{a}$

For $x >> a$, $V = \frac{kq}{x}$

(Like that of a point charge).

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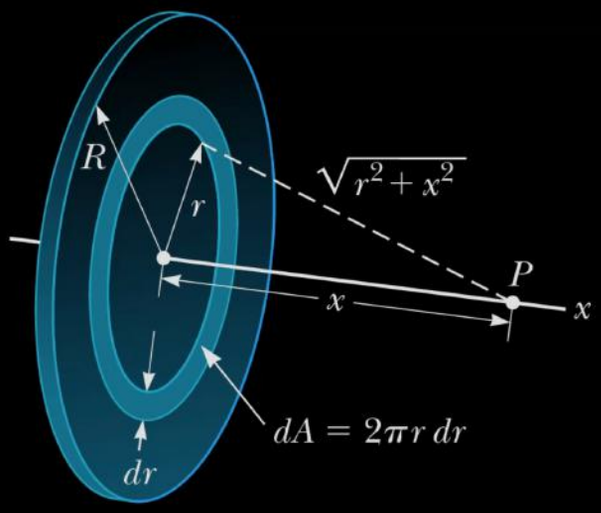
Potential Due To A Disc:

Field due to a disc can be found by breaking it in to several ring elements:

$dV = k\,dq / \sqrt{x^2+r^2}$, $dq = \sigma \times 2\pi r\,dr$

$$\Rightarrow V = \pi k\sigma \int_0^R \frac{2r\,dr}{\sqrt{r^2+a^2}}$$

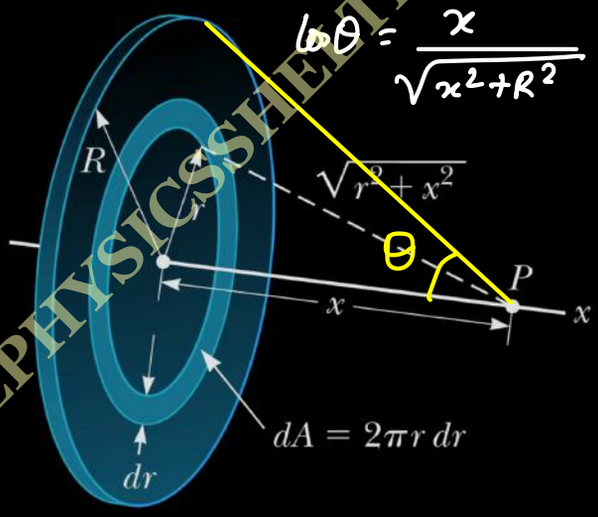
$$\Rightarrow \boxed{V = 2\pi k\sigma\left[\sqrt{R^2+x^2} - x\right]}$$



Alternatively Recall that normal field is given as $\sigma k\omega$, thus for disc field is $k\sigma \times 2\pi(1-\cos\theta)$ $\Rightarrow$

$$E = 2\pi k\sigma\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right] = -\frac{dV}{dx}$$

$$\cos\theta = \frac{x}{\sqrt{x^2+R^2}}$$



Now Integrating,

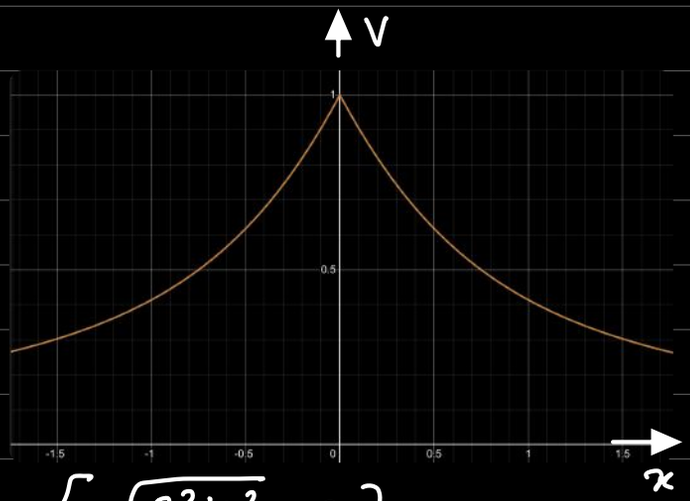
$V = 2\pi k\sigma\left[\sqrt{R^2+x^2} - x\right] + C$,

Taking zero potential at ∞, $C = 0$.

Thus $V = 2\pi k\sigma\left[\sqrt{R^2+x^2} - x\right]$

Plot of Potential due to a disc:

Maximum at centre and goes to zero as $x \to \infty$. For a disc also it can be shown that at very large distances, potential formula reduces to that of point charge.



$\sigma = Q/\pi R^2$ $\Rightarrow$ $V = 2\pi k \frac{Q}{\pi R^2}\left[\sqrt{R^2+x^2} - x\right]$

$$\Rightarrow V = \frac{2kQx}{R^2}\left[\sqrt{1+(R/x)^2} - 1\right]$$, If $x >> 1$, Applying Binomial,

$$V = \frac{2kQx}{R^2}\left[1 + \frac{1}{2}\frac{R^2}{x^2} - 1\right] = \frac{kQ}{x}$$ ← Just like a point charge.

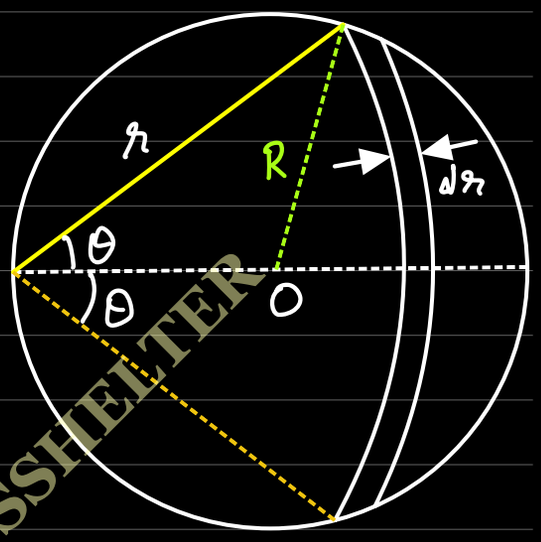
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→ For distances much larger than dimensions of the charge distribution the field and potential of every distribution would start approaching that of a point charge. [Try verifying this for ring also].

Potential At Periphery Of Uniformly Charged Disc:

Consider an arc element of a uniformly charged disc. The arc is centred at point P. Since all points of arc are equidistant from P, its potential contribution is given by:



$$dV = \frac{k\,dq}{r}, \quad dq = \sigma \times 2\pi\theta\, dr$$

$$\Rightarrow dV = \frac{k}{r}\cdot \sigma \times 2\pi\theta\, dr = 2k\sigma\theta\, dr$$

Now $r = 2R\cos\theta \Rightarrow dr = -2R\sin\theta\, d\theta$

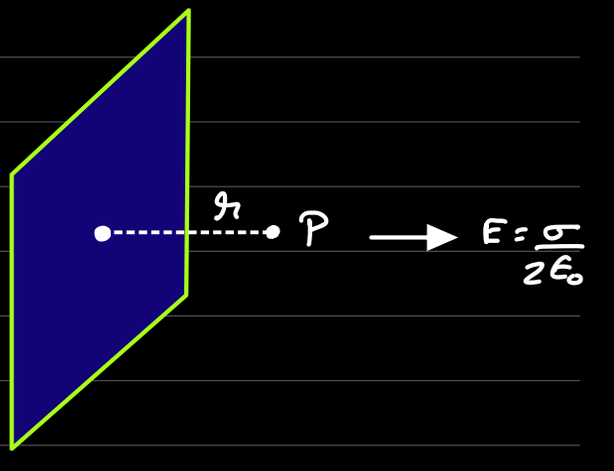
$$\Rightarrow dV = 2\times\frac{\sigma}{4\pi\epsilon_0}\times\theta\times(-2R)\sin\theta\, d\theta = -\frac{\sigma R}{\pi\epsilon_0}\theta\sin\theta\, d\theta$$

$$\Rightarrow V = -\frac{\sigma R}{\pi\epsilon_0}\int_0^{\pi/2}\theta\sin\theta\, d\theta$$, Integrating by parts, we get:

$$\boxed{V = \sigma R/\pi\epsilon_0}$$

Potential Due To Infinite Uniform Sheet:

Recall that normal field due to an infinite lamina is given by $E = k\sigma\omega$, now due to an infinite sheet, at any finite point in space, solid angle is 2π steradians (For source of light very near the plane, half of spherical screen will be in shadow). Thus field due to an infinite sheet is:



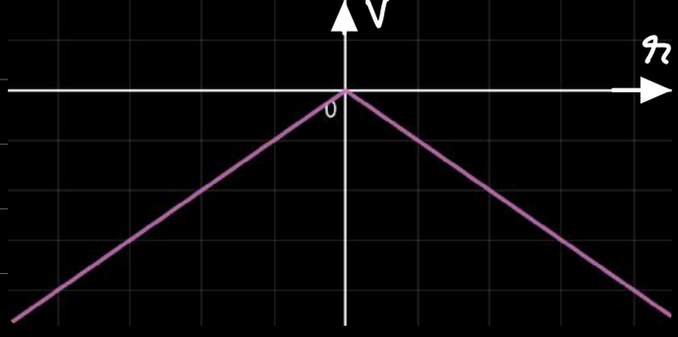
$$E = \frac{1}{4\pi\epsilon_0}\times\sigma\times 2\pi = \frac{\sigma}{2\epsilon_0} = -\frac{dV}{dr}$$

$$\Rightarrow V = -\frac{\sigma r}{2\epsilon_0} + C$$, Now taking reference potential 0 at $r = 0$,

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For infinite sheet, $\boxed{V = -\frac{\sigma r}{2\epsilon_0}}$

POTENTIAL Vs r GRAPH FOR ∞ SHEET →



→ Note that here it is not convenient to assume zero potential at ∞ because, that would make $C = \infty$ and thus potential at every point will become ∞ and for finding potential differences we will have to deal with $\infty - \infty$ forms which is absurd.

POTENTIAL DUE TO INFINITE LINE CHARGE :

Recall that field due to an infinite line charge is given as:

$$E = \frac{2k\lambda}{r} = \frac{\lambda}{2\pi\epsilon_0 r} = -\frac{dV}{dr}$$

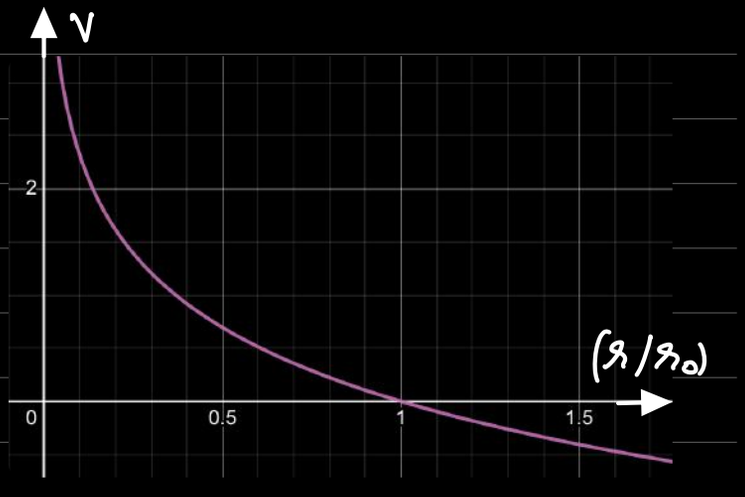


$$\Rightarrow V = \frac{-\lambda}{2\pi\epsilon_0}\int\frac{dr}{r} = \frac{-\lambda}{2\pi\epsilon_0}\ln(r) + C$$

If we assume zero potential at $r = r_0$, $C = \frac{\lambda}{2\pi\epsilon_0}\ln(r_0)$

Thus $\boxed{V = \frac{\lambda}{2\pi\epsilon_0}\ln(r_0/r)}$

Plot of V Vs r/r_0 →



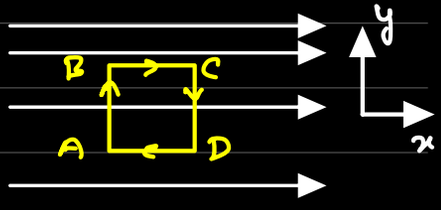
→ Here also choosing zero potential at ∞ or $r = 0$ would be inconvenient as $\ln(\infty)$ as well as $\ln(0)$ are not defined. Therefore we chose zero potential at some finite distance r_0 from wire. Now we can conveniently write potential difference between two points as $V_2 - V_1 = \frac{\lambda}{2\pi\epsilon_0}\ln(r_1/r_2)$

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EXAMPLE: Can shown Electric field lines represent an electrostatic field?



Solution: Consider the closed Path ABCDA.
Since the field is // but not uniform, work done along BC is greater in magnitude than the work done along AD. Also work done along AB & CD is zero ∴ work done is closed loop is non zero ∴ field is non conservative and hence cannot represent electrostatic field. Alternately, $E_y = 0$ (identically)
∴ $\frac{\partial E_y}{\partial x} = 0$ However $\frac{\partial E_x}{\partial y} \neq 0$ ∴ field can't be conservative.

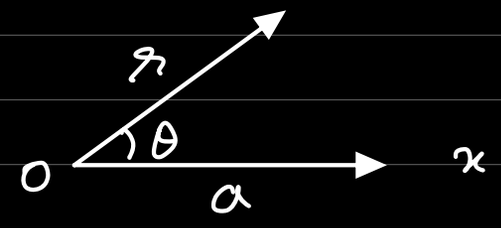


[Recall that for conservative fields, cross derivatives are equal].

EXAMPLE: IRODOV 3.35

Find the electric field strength vector if the potential of this field has the form $\phi = \vec{a}\cdot\vec{r}$, where $\vec{a}$ is a constant vector, and $\vec{r}$ is the radius vector of a point of the field.

Solution: Without loss of generality let us take x axis to coincide with $\vec{a}$.



Now $\phi = \vec{a}\cdot\vec{r} = ar\cos\theta = ax$

Now $E_x = -\partial\phi/\partial x = -a$

also $E_y = E_z = 0$ ($\because \frac{\partial\phi}{\partial y} = \frac{\partial\phi}{\partial z} = 0$).

Thus $\vec{E} = -\vec{a}$ Ans.

EXAMPLE: IRODOV 3.37:

The potential of a certain electrostatic field has the form $\phi = a(x^2 + y^2) + bz^2$, where a and b are constants. Find the magnitude and direction of the electric field strength vector. What shape have the equipotential surfaces in the following cases:
(a) $a > 0, b > 0$; (b) $a > 0, b < 0$?

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Solution: $\vec{E} = -\nabla\phi = -[2ax\hat{i} + 2ay\hat{j} + 2bz\hat{k}]$

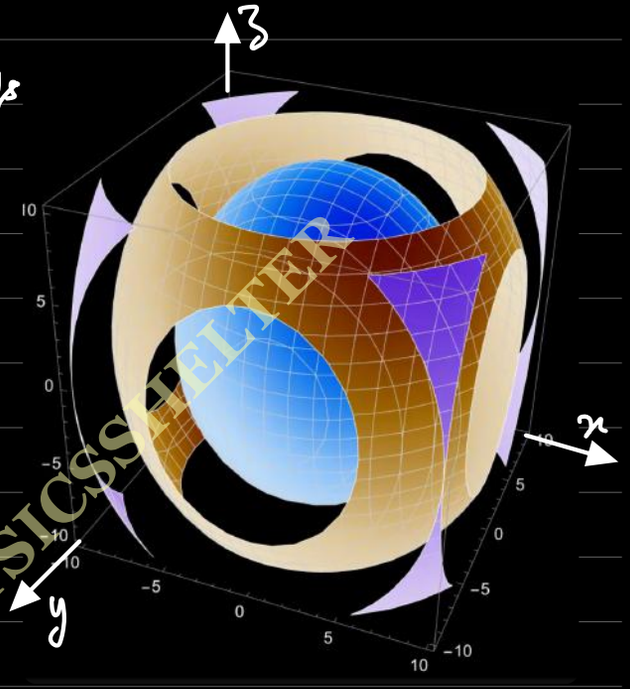
If $a>0$ & $b>0$, $\phi = a(x^2+y^2) + bz^2 =$ Constant

$\Rightarrow$ $ax^2 + bz^2 = C - ay^2 \Rightarrow$ x-z projection is ellipse

also $ay^2 + bz^2 = C - ax^2 \Rightarrow$ y-z projection is ellipse

also $ax^2 + ay^2 = C - bz^2 \Rightarrow$ x-y projection is circular.

Thus the equipotential surfaces are ellipsoids with xy cross sections being circular y-z and z-x cross sections being ellipses. (See adjacent figure).



→ General equation of ellipsoid is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

(b) If $a>0$, $b<0$

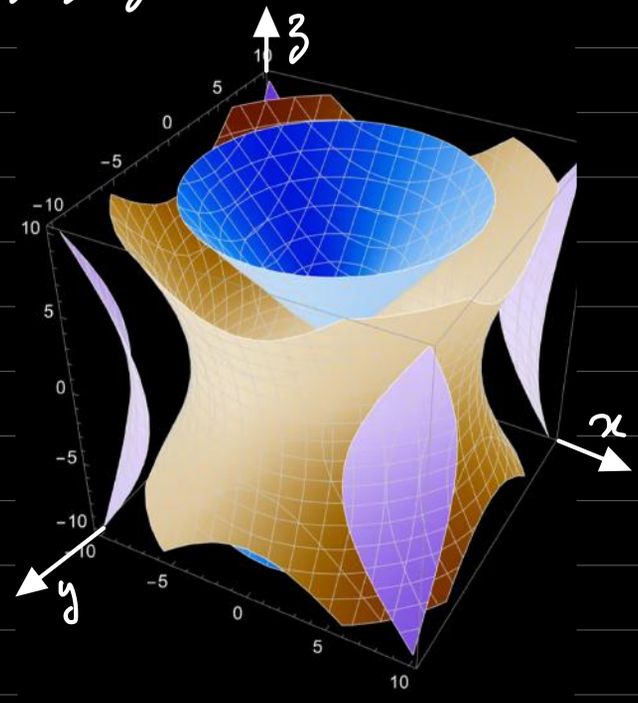
$ax^2 + bz^2 = C - ay^2 \Rightarrow$ xz projection is hyperbolic

also $ay^2 + bz^2 = C - ax^2 \Rightarrow$ yz projections is hyperbolic

$ax^2 + ay^2 = C - bz^2 \Rightarrow$ xy projection is circular.

Refer to the adjacent figure, one can clearly see that required surface is a hyperboloid. (Surface of revolution of a hyperbola about z axis).

In general any equation of the form $ax^2 + by^2 + cy^2 = C$ represents a conicoid (paraboloid / hyperboloid / Ellipsoid, Sphere etc). Such conicoids will have their main axes along x, y & z axes just as the ellipse $\frac{x^2}{a} + \frac{y^2}{b^2} = 1$ has its major and minor axes along coordinate axes.



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Example: PE of a system with all charges in equilibrium (Kevin Zhou)

Prove that when a system of point charges is in equilibrium (i.e. the net force on *each* of the charges due to the others vanishes), the total potential energy of the system is zero.

Solution: If each charge is in equilibrium, then force on each charge must be zero ∴ even if the system's dimensions are scaled up by some constant factor, the work done must be zero (constant could be arbitrarily big). By choosing arbitrarily big scaling factor, charges will be at infinite separation and ∴ system will have zero potential energy. However ∵ electric field didn't do any work ∴ PE must have been zero in the first place.

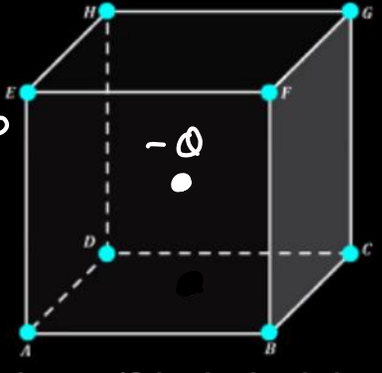
Example (Kevin Zhou): Always Decreasing Potential :

Show that for a positive point charge in the electric fields of fixed, positive point charges, there is a path along which the charge can be moved to infinity without ever needing positive external work, i.e. a path along which the potential only decreases.

Solution: Consider a field line going through the test charge, Since there is no negative charge anywhere in space ∴ this field can't terminate anywhere except at ∞. Thus there exists a field line all the way up to infinity and this is our required path. (This is somewhat related to Earnshaw's theorem about which we shall learn after learning Gauss' Law).

Example: Eight charges each of magnitude q are fixed at vertices of a cube, what charge $-Q$ must be placed at the centre of the cube so that each charge is in equilibrium.

Solution: As learnt in an example above, the net electrostatic energy of the system must be zero.



$$\Rightarrow \underbrace{(kq^2/a)\times 12}_{\text{Edges}} + \underbrace{(kq^2/a\sqrt{2})\times 12}_{\text{Face diag.}} + \underbrace{(kq^2/a\sqrt{3})\times 4}_{\text{Body Diag.}} - \underbrace{kQq/(a\sqrt{3}/2)=0}_{\text{Centre}} \quad \times 8$$

$$\Rightarrow Q = q[2+6\sqrt{3}+3\sqrt{6}] \text{ Ans.}$$

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